

Region Growing & Region Splitting

Region-based segmentation is based on the connectivity of similar pixels in a region

Two main approaches to region-based segmentation

① Region-growing

② Region-splitting

Procedure for Region-Growing

Q1.

	0	1	2	3
0	0	1	2	0
1	2	5	6	0
2	1	4	7	3
3	0	2	5	1

Apply region-growing on the following image with initial point (2,2) and threshold value as 2. Use 4 connectivity

$$T=2$$

$$|7-6| = 1 \leq 2 \quad \checkmark$$

$$|7-4| = 3 \not\leq 2 \quad \times$$

$$|7-5| = 2 \leq 2 \quad \checkmark$$

$$|7-3| = 4 \not\leq 2 \quad \times$$

$$|Seed\ value - Pixel\ value| \leq Threshold$$

Same Region

Otherwise, Separate Region

Q1 If specific point given, region growed step-by-step

Q2 Q3 If intensity value given as seed point, region growed ~~once~~ step-by step also

↓

0	1	2	0
2	5	6	0
1	4	7	3
0	2	5	1

$$|6-2|=4 \not\leq 2$$

$$|5-2|=3 \not\leq 2$$

$$|6-5|=1 \leq 2$$

$$|5-1|=4 \not\leq 2$$

$$|6-0|=6 \not\leq 2$$

Note: Place X where condition is NOT satisfied

↓

0	1	2	0
2	5	6	0
1	4	7	3
0	2	5	1

$$|5-1|=4 \not\leq 2$$

$$|5-2|=3 \not\leq 2$$

$$|5-4|=1 \leq 2$$

[This satisfies for 5, but not for 7]
no need to even calculate

Note: If even one X, DO NOT MERGE

Final Result (Segmented Image)

0	0	0	0
0	1	1	0
0	0	1	0
0	0	1	0

If connectivity not mentioned, 8 way connectivity
 * Max difference between pixels in a region must be $\leq$ Threshold

Q2.

5	6	6	7	6	7	6	6
6	7	6	7	5	5	4	7
6	6	4	4	3	2	5	6
5	4	5	4	2	3	4	6
0	3	2	3	3	2	4	7
0	0	0	0	2	2	5	6
1	1	0	1	0	3	4	4
1	0	1	0	2	3	5	4

Apply region-growing
 on following image
 with seed point
 as 6 (intensity value)
 and threshold value
 as 3.

Final Result (Segmented Image)

1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1
1	1	1	1	0	0	1	1
1	1	1	1	0	0	1	1
0	0	0	0	0	0	1	1
0	0	0	0	0	0	1	1
0	0	0	0	0	0	1	1
0	0	0	0	0	0	1	1

① Put 1's where
 initial
 seed points (6) are

② Shortcut:

$$6 - 3 = 3$$

Put 1's where
 intensity value
 3 or higher (≥ 3)
 and is 8-neighbors
 with 6's

③ The 1's put in step-2
 are the new seed points

Check with neighbors, if
 threshold condition satisfied, put 1's in neighbor

*
 * ④ Note: Max difference between pixels in a region
 must be $\leq$ Threshold

threshold condition satisfied, put 1's in neighbor

Q3.

2	2	7	2	1
1	7	6	6	2
7	6	6	5	7
2	4	5	4	2
1	2	5	1	1

Seed Point = 6

Threshold = 3

How many segments identified?

Segmented Image

0	0	1	0	0
0	1	1	1	0
1	1	1	1	1
0	1	1	1	0
0	0	1	0	0

5 segments

Region Splitting

Q4

5	6	6	6	7	7	6	6
6	7	6	7	5	5	4	7
6	6	4	4	3	2	5	6
5	4	5	4	2	3	4	6
0	3	2	3	3	2	4	7
0	0	0	0	2	2	5	6
1	1	0	1	0	3	4	4
1	0	1	0	2	3	5	4

Apply region-splitting on the following image. Assume the threshold value be ≤ 4

i.e. $|Max - Min| \leq 4$ [Don't Split]

$|Max - Min| > 4$ [Split]

Max Intensity Level = 7

Min Intensity Level = 0

$7 - 0 > 4$ [$\therefore$ Split]

(a)

$7 - 4 \leq 4$

5	6	6	6	7	7	6	6
6	7	6	7	5	5	4	7
6	6	4	4	3	2	5	6
5	4	5	4	2	3	4	6
0	3	2	3	3	2	4	7
0	0	0	0	2	2	5	6
1	1	0	1	0	3	4	4
1	0	1	0	2	3	5	4

$3 - 0 \leq 4$

(c)

(b)

$7 - 2 > 4$ $\rightarrow$ Split

$7 - 0 > 4$ $\rightarrow$ Split

(d)

Image divided into a, b, c, d regions. Then, repeat process for each region

a

5	6	6	6	7	7	6	6
6	7	6	7	5	5	4	7
6	6	4	4	3	2	5	6
5	4	5	4	2	3	4	6
0	3	2	3	3	2	4	7
0	0	0	0	2	2	5	6
1	1	0	1	0	3	4	4
1	0	1	0	2	3	5	4

b₁

b₂

$$7-5 \leq 4 \checkmark$$

$$7-4 \leq 4 \checkmark$$

$$3-2 \leq 4 \checkmark$$

$$6-5 \leq 4 \checkmark$$

$$3-2 \leq 4 \checkmark$$

$$7-4 \leq 4 \checkmark$$

$$3-0 \leq 4 \checkmark$$

$$5-4 \leq 4 \checkmark$$

No More Splitting
Required

d₃

d₄

10 Regions

□ Merging

After region-splitting, check adjacent regions. (8 connectivity)

If in the union of two regions,

$$\text{Max} - \text{Min} \leq \text{Threshold} \quad [\text{Merge}]$$

$$\text{Max} - \text{Min} \not\leq \text{Threshold} \quad [\text{Don't Merge}]$$

5	6	6	6	7	7	6	6
6	7	6	7	5	5	4	7
6	6	4	4	3	2	5	6
5	4	5	4	2	3	4	6
0	3	2	3	3	2	4	7
0	0	0	0	2	2	5	6
1	1	0	1	0	3	4	4
1	0	1	0	2	3	5	4

Q. Apply splitting and merging,
on the image with threshold value
equal to 3

Step-1: Splitting
Condition

$$\text{Max value} - \text{Min value} \leq 3$$

$$7 - 0 \not\leq 3$$

(a)	5	6	6	6	7	7	6	6	(b)
	6	7	6	7	5	5	4	7	
	6	6	4	4	3	2	5	6	
	5	4	5	4	2	3	4	6	
	0	3	2	3	3	2	5	7	
	0	0	0	0	2	2	5	6	
	1	1	0	1	0	3	4	4	
	1	0	1	0	2	3	5	4	

a-region

$$7-4 \leq 3$$

b-region

$$7-2 \not\leq 3$$

c-region

$$3-0 \leq 3$$

d-region

$$7-0 \not\leq 3$$

$\therefore$ Regions b and d split into 4 sub-regions

(a)	5	6	6	6	(b ₁)	7	7	6	6	(b ₂)
	6	7	6	7		5	5	4	7	
	6	6	4	4		3	2	5	6	
	5	4	5	4	(b ₃)	2	3	4	6	(b ₄)
	0	3	2	3	(d ₁)	3	2	5	7	(d ₂)
	0	0	0	0		2	2	5	6	
	1	1	0	1		0	3	4	4	
	1	0	1	0		2	3	5	4	

b₁-region

$$7-5 \leq 3$$

b₂-region

$$7-4 \leq 3$$

b₃-region

$$3-2 \leq 3$$

b₄-region

$$6-4 \leq 3$$

d₁-region

$$3-2 \leq 3$$

d₂-region

$$7-5 \leq 3$$

d₃-region

$$3-0 \leq 3$$

d₄-region

$$5-4 \leq 3$$

Step-2: Merging

a				b ₁		b ₂	
5	6	6	6	7	7	6	6
6	7	6	7	5	5	4	7
6	6	4	4	3	2	5	6
5	4	5	4	2	3	4	6
0	3	2	3	3	2	5	7
0	0	0	0	2	2	5	6
1	1	0	1	0	3	4	4
1	0	1	6	2	3	5	4
c				d ₃		d ₄	

① If in two adjacent regions (8-neighbors),

$$\text{Max value} - \text{Min value} \leq \text{Threshold}$$

then Merge

② However, the condition (Max value - Min value $\leq$ Threshold) must be satisfied by other regions in the merger as well

Threshold = 3

① Regions a and b₁

$$7 - 4 \leq 3$$

$\therefore$ Regions merged

② Regions a and b₃

$$7 - 2 \not\leq 3$$

③ Regions b₁ and b₂

$$7 - 4 \leq 3$$

$\therefore$ Regions merged

④ Regions b₂ and b₄

$$7 - 4 \leq 3$$

$\therefore$ Regions merged

⑤ Regions b₄ and d₂

$$7 - 4 \leq 3$$

$\therefore$ Regions merged

⑥ Regions d₂ and d₁

$$7 - 2 \not\leq 3$$

⑦ Regions d₂ and d₄

$$7 - 4 \leq 3$$

$\therefore$ Regions merged

⑧ Regions d₄ and d₃

$$4 - 0 \not\leq 3$$

$\therefore$ Regions a, b₁, b₂, b₄, d₂, d₄, merged into one region

Note: Regions d₄ and d₁ satisfy the threshold condition. However, d₂ has already been selected and merged, and d₂ and d₁ do not satisfy threshold

Note: If not specified, start merging from region at (0,0)

① Regions b_3 and d_1

$$3-2 \leq 3$$

$\therefore$ Regions merged

② Regions b_3-d_1 and d_3

$$3-0 \leq 3$$

$\therefore$ Regions merged

③ Regions $b_3-d_1-d_3$ and d_1

$$3-0 \leq 3$$

Regions merged

$\therefore$ Regions b_3, d_1, d_3, c are merged into one region

①

5	6	6	6	7	7	6	6
6	7	6	7	5	5	4	7
6	6	4	4	3	2	5	6
5	4	4	4	2	3	4	6
0	3	2	3	3	2	5	7
0	0	0	0	2	2	5	6
1	1	0	1	0	3	4	4
1	0	1	0	2	3	5	4

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Note: Different answers possible depending on which region considered next at each step